Principles of Communications - CIA 1 Answer Key

Set A

Part-A

1. Draw the block diagram of communication systems.

A communication system consists of five key elements: **Information Source** → **Transmitter** → **Channel** → **Receiver** → **Destination**, with noise entering at the channel stage.

```
                              Noise
                                |
[Info Source] → [Transmitter] → [Channel] → [Receiver] → [Destination]
                       ↑                            |
                [Modulator,                  [Demodulator,
                 Encoder]                      Decoder]
```

The transmitter modulates the message onto a carrier for efficient transmission. The channel (wired/wireless) introduces attenuation and noise. The receiver recovers the original signal through demodulation and decoding.

2. Compare DSB-SC and SSB.

Parameter	DSB-SC	SSB
Sidebands transmitted	Both USB and LSB	Only one (USB or LSB)
Carrier	Suppressed	Suppressed
Bandwidth	$2f_m$	f_m
Power efficiency	Moderate	Best (all power in sideband)
Demodulation	Coherent detection required	Coherent detection required
Circuit complexity	Moderate	High (sharp BPF or phase-shift method)
Application	Data links, pilot-carrier systems	HF voice, amateur radio

3. For an FM system, the maximum deviation is 75 kHz and the highest modulating frequency is 15 kHz. Determine the required transmission bandwidth using Carson's rule.

Carson's rule:

$$B_T = 2(\Delta f + f_m)$$

$$B_T = 2(75 + 15) = 2 \times 90 = \boxed{180\ \text{kHz}}$$

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4. Identify suitable modulation technique for TV transmission and justify.

VSB (Vestigial Sideband) Modulation is used for TV video transmission. TV video signals have baseband content from DC to ~4 MHz. SSB would require an ideal sharp-cutoff filter, which is impractical near DC. DSB would require 8 MHz bandwidth, exceeding the 6 MHz channel allocation. VSB transmits the full upper sideband (4 MHz) plus a vestige (~1.25 MHz) of the lower sideband, fitting within 6 MHz while allowing simpler demodulation with minimal low-frequency distortion.

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5. Classify the FM signal as NBFM or WBFM if β = 0.5.

Since $\beta = 0.5 < 1$, the signal is **Narrowband FM (NBFM)**.

NBFM criterion: $\beta \leq 1$ radian. At $\beta = 0.5$, only the carrier and two significant sidebands ($f_c \pm f_m$) are present, giving bandwidth $\approx 2f_m$, similar to AM. The phase deviation is small enough to use the small-angle approximation.

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6. Summarize the key differences between a random variable and a random process.

A **random variable** maps a single experimental outcome to one real number - it is a fixed (though uncertain) value. A **random process** is an indexed family of random variables over time $\{X(t), t \in T\}$; each outcome maps to an entire waveform (time function). A random variable is a snapshot; a random process describes a signal that evolves randomly over time. Noise in a communication receiver is modeled as a random process, whereas a single amplitude measurement is a random variable.

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7. Discuss the mathematical conditions for Wide Sense Stationary.

A random process $X(t)$ is **Wide Sense Stationary (WSS)** if:

Condition 1 - Constant Mean:

$$E[X(t)] = \mu_X \quad \text{(constant, independent of } t\text{)}$$

Condition 2 - Autocorrelation depends only on time difference:

$$R_X(t_1, t_2) = E[X(t_1)X(t_2)] = R_X(\tau), \quad \tau = t_1 - t_2$$

These are weaker than the SSS conditions. WSS is sufficient for spectral analysis via the Wiener-Khinchin theorem: $S_X(f) = \mathcal{F}\{R_X(\tau)\}$.

---

8. Show how narrowband noise behaves in a modulated system and state its characteristics.

Narrowband noise (noise passed through a bandpass filter centered at f_c) is represented as:

$$n(t) = n_I(t)\cos(2\pi f_c t) - n_Q(t)\sin(2\pi f_c t)$$

Characteristics:

- $n_I(t)$ (in-phase) and $n_Q(t)$ (quadrature) are lowpass processes with the same PSD and power as $n(t)$
- They are mutually uncorrelated: $E[n_I(t)n_Q(t)] = 0$
- In AM envelope detection, only $n_I(t)$ appears at output (at high SNR)
- In FM demodulation, $n_Q(t)$ dominates and its PSD grows parabolically with frequency

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9. Apply the noise equivalent bandwidth formula to analyze receiver performance.

The noise equivalent bandwidth B_{eq} replaces a real filter with an ideal rectangular filter of the same noise output power:

$$B_{eq} = \frac{\int_0^\infty |H(f)|^2 \, df}{|H(f_{max})|^2}$$

Output noise power: $N_o = N_0 \cdot |H_{max}|^2 \cdot B_{eq}$

For a single-pole RC filter with $H(f) = \frac{1}{1+j(f/f_3)}$: $B_{eq} = \frac{\pi}{2} f_3 \approx 1.57 f_3$, which is wider than the 3 dB bandwidth. This shows that the actual noise passed is more than what the 3 dB bandwidth alone would suggest, which is critical for accurate receiver SNR calculations.

Part-B

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i. Demonstrate how an AM signal is generated using a square law modulator and derive its mathematical equation.

Principle:

A square-law modulator uses a nonlinear device (diode or FET biased in the square-law region) whose transfer characteristic approximates:

$$v_o = a_1 v_i + a_2 v_i^2$$

Block Diagram:

```
m(t) ──┐
       ├──[Σ]──── v_i ──[Square-Law Device]──[BPF @ f_c]──→ s_AM(t)
c(t) ──┘                v_o = a₁vᵢ + a₂vᵢ²
= Ac·cos(ωct)
```

Derivation:

The input to the device:

$$v_i(t) = A_c \cos \omega_c t + m(t)$$

Applying the square-law characteristic:

$$v_o = a_1[A_c \cos \omega_c t + m(t)] + a_2[A_c \cos \omega_c t + m(t)]^2$$

Expanding the squared term:

$$v_o = a_1 A_c \cos \omega_c t + a_1 m(t) + a_2 A_c^2 \cos^2 \omega_c t + 2a_2 A_c m(t) \cos \omega_c t + a_2 m^2(t)$$

Using $\cos^2 \omega_c t = \frac{1}{2}(1 + \cos 2\omega_c t)$:

$$v_o = \underbrace{a_1 m(t) + a_2 m^2(t) + \frac{a_2 A_c^2}{2}}_{\text{lowpass terms}} + \underbrace{[a_1 A_c + 2a_2 A_c m(t)] \cos \omega_c t}_{\text{AM term at } f_c} + \underbrace{\frac{a_2 A_c^2}{2} \cos 2\omega_c t}_{\text{at } 2f_c}$$

After the BPF centered at f_c passes only the term at ω_c:

$$\boxed{s_{AM}(t) = a_1 A_c \left[1 + \frac{2a_2}{a_1} m(t)\right] \cos \omega_c t}$$

This is the standard AM signal with modulation sensitivity $k_a = \dfrac{2a_2}{a_1}$.

For a single-tone message $m(t) = A_m \cos \omega_m t$, the modulation index is:

$$\mu = k_a A_m = \frac{2a_2 A_m}{a_1}$$

Spectrum of the output:

```
S(f)
  |
  |    (carrier)     LSB        USB
  |        |          |          |
  |        ↓          ↓          ↓
──┼────────┬──────────┬──────────┬────→ f
         fc-fm       fc        fc+fm
```

The distortion term $a_2 m^2(t)$ is removed by the BPF. The output power is:

$$P_{AM} = \frac{(a_1 A_c)^2}{2}\left(1 + \frac{\mu^2}{2}\right)$$

Limitation: The square-law approximation is valid only for small signal swings. For larger signals, higher-order nonlinear terms introduce additional harmonic distortion components that must be filtered out.

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ii. Construct the block diagram based on Weaver's method and describe how it generates an SSB signal.

Weaver's Method (Third Method):

Weaver's method generates SSB using two stages of modulation and lowpass filters, avoiding the need for an ideal sharp-cutoff sideband filter.

Block Diagram:

```
                    ┌─[×]──[LPF, BW=fm/2]──[×]──────┐
                    │   ↑                    ↑       │
m(t) ───────[S]───┤  cos(ω₀t)            cos(ωct)  ├──[Σ]──→ USB / LSB
                    │                                 │  (+ for USB)
                    │   ↑                    ↑       │  (- for LSB)
                    └─[×]──[LPF, BW=fm/2]──[×]──────┘
                      sin(ω₀t)             sin(ωct)
```

where f_0 = midband frequency of message (typically $\frac{f_{m,max}+f_{m,min}}{2}$, e.g., 2 kHz for audio 300 Hz - 3.4 kHz).

Step-by-step mathematical analysis for $m(t) = A_m \cos \omega_m t$:

Upper path:

After first multiplier:

$$u_1(t) = A_m \cos\omega_m t \cdot \cos\omega_0 t = \frac{A_m}{2}[\cos(\omega_m - \omega_0)t + \cos(\omega_m + \omega_0)t]$$

After LPF (retains $\omega_m - \omega_0$ component, since $|\omega_m - \omega_0| < \omega_0$):

$$u_2(t) = \frac{A_m}{2}\cos(\omega_m - \omega_0)t$$

After second multiplier with $\cos\omega_c t$:

$$u_3(t) = \frac{A_m}{4}[\cos(\omega_c + \omega_m - \omega_0)t + \cos(\omega_c - \omega_m + \omega_0)t]$$

Lower path:

After first multiplier with $\sin\omega_0 t$:

$$l_1(t) = \frac{A_m}{2}[\sin(\omega_m - \omega_0)t + \sin(\omega_m + \omega_0)t]$$

After LPF:

$$l_2(t) = \frac{A_m}{2}\sin(\omega_m - \omega_0)t$$

After second multiplier with $\sin\omega_c t$:

$$l_3(t) = \frac{A_m}{4}[\cos(\omega_c - \omega_m + \omega_0)t - \cos(\omega_c + \omega_m - \omega_0)t]$$

Adding (USB):

$$s_{USB}(t) = u_3 + l_3 = \frac{A_m}{2}\cos(\omega_c + \omega_m - \omega_0)t$$

Subtracting (LSB):

$$s_{LSB}(t) = u_3 - l_3 = \frac{A_m}{2}\cos(\omega_c - \omega_m + \omega_0)t$$

Advantages:

- Avoids sharp-cutoff filters - LPFs are easy to build
- Requires only a 90° phase shift at fixed frequency f_0, not across the whole signal band
- Well-suited for digital/DSP implementation

Disadvantage: Any imbalance in the two paths introduces the unwanted sideband (carrier/sideband suppression depends on component matching). The DC offset at the second LPF output also creates a tone at $f_c - f_0$ if not carefully handled.

i. Apply angle modulation principles to formulate the mathematical expressions of FM and PM waves.

General Angle Modulated Signal:

$$s(t) = A_c \cos[\omega_c t + \phi(t)]$$

where $\phi(t)$ is the instantaneous phase deviation. The instantaneous frequency is:

$$f_i(t) = f_c + \frac{1}{2\pi}\frac{d\phi(t)}{dt}$$

Phase Modulation (PM):

The instantaneous phase varies linearly with the message:

$$\phi_{PM}(t) = k_p \cdot m(t)$$

$$\boxed{s_{PM}(t) = A_c \cos[\omega_c t + k_p m(t)]}$$

For single-tone $m(t) = A_m \cos \omega_m t$:

$$s_{PM}(t) = A_c \cos[\omega_c t + k_p A_m \cos \omega_m t] = A_c \cos[\omega_c t + \beta_p \cos \omega_m t]$$

where $\beta_p = k_p A_m$ = PM modulation index (rad).

Instantaneous frequency of PM:

$$f_i(t) = f_c - k_p A_m f_m \sin \omega_m t = f_c - \beta_p f_m \sin \omega_m t$$

Maximum frequency deviation: $\Delta f_{PM} = k_p A_m f_m$ (increases with f_m)

Frequency Modulation (FM):

The instantaneous frequency varies linearly with the message:

$$f_i(t) = f_c + k_f m(t)$$

Integrating to get phase:

$$\phi_{FM}(t) = 2\pi k_f \int_0^t m(\tau)\, d\tau$$

$$\boxed{s_{FM}(t) = A_c \cos\left[2\pi f_c t + 2\pi k_f \int_0^t m(\tau)\, d\tau\right]}$$

For single-tone $m(t) = A_m \cos \omega_m t$:

$$\phi_{FM}(t) = \frac{k_f A_m}{f_m} \sin \omega_m t = \beta_f \sin \omega_m t$$

$$\boxed{s_{FM}(t) = A_c \cos[\omega_c t + \beta_f \sin \omega_m t]}$$

where $\beta_f = \frac{\Delta f}{f_m} = \frac{k_f A_m}{f_m}$ = FM modulation index.

Maximum frequency deviation: $\Delta f = k_f A_m$ (independent of f_m)

Bessel Function Expansion of FM:

Using the identity $e^{j\beta \sin\theta} = \sum_{n=-\infty}^{\infty} J_n(\beta) e^{jn\theta}$:

$$s_{FM}(t) = A_c \sum_{n=-\infty}^{\infty} J_n(\beta) \cos(\omega_c + n\omega_m)t$$

The spectrum contains sidebands at $f_c \pm n f_m$ with amplitudes $A_c J_n(\beta)$. Total power remains $A_c^2/2$ regardless of β.

Carson's Rule Bandwidth:

$$B_T = 2(\Delta f + f_m) = 2f_m(1 + \beta)$$

FM vs PM Comparison:

Property	FM	PM
$\phi(t)$	$\propto \int m(t)dt$	$\propto m(t)$
$f_i(t)$	$\propto m(t)$	$\propto \dot{m}(t)$
β vs A_m	Proportional	Proportional
β vs f_m	Inversely proportional	Independent
Δf vs f_m	Independent	Proportional

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ii. Illustrate the process of FM demodulation with balanced slope detector with the circuit diagram and characteristic curve.

Principle:

The balanced slope detector (Travis discriminator) converts frequency variations to amplitude variations, which are then envelope-detected. Two tuned circuits detuned above and below f_c produce complementary amplitude responses, giving a linear output vs frequency characteristic.

Circuit Diagram:

```
                 L1             C1
     ┌──────UUUU──────||─────┬───[D1]──┐
```

```
                 │    (tuned to           │          │
FM Input ────────┤     fc + Δf)           │        [R1]───┐
(via              │                      │          │     │
 coupling)       │     L2          C2   │         [Ca]    │
                 └─────UUUU──────────||──────┴───[D2]───┘     ├── Vout
                       (tuned to                   │          │
                        fc - Δf)                [R2]───┐     │
                                                 │      │     │
                                                [Cb]   └─────┘
                                                 │
                                                GND
```

Two tuned circuits with opposite detuning. Diodes D1 and D2 are connected in opposing polarity.

Operation Analysis:

At **carrier frequency** $f = f_c$:

- Both circuits equally detuned → equal envelope outputs
- $V_{D1} = V_{D2} \rightarrow V_{out} = V_{D1} - V_{D2} = 0$

At $f > f_c$ (positive deviation):

- L1C1 (tuned higher) moves toward resonance → $|V_{D1}|$ increases
- L2C2 (tuned lower) moves away from resonance → $|V_{D2}|$ decreases
- $V_{out} = V_{D1} - V_{D2} > 0$ → Positive output

At $f < f_c$ (negative deviation):

- L1C1 moves further from resonance → $|V_{D1}|$ decreases
- L2C2 moves toward resonance → $|V_{D2}|$ increases
- $V_{out} = V_{D1} - V_{D2} < 0$ → Negative output

Characteristic S-Curve:

```
Output
Voltage
(V)
 +Vmax |           ___________
       |          /
       |         /
   0   |--------/--------------→  Frequency (f)
       |       /    fc
       |      /
 -Vmax|_____/
       |
```

```
        ← Linear region →
        (fc - Δf)  to  (fc + Δf)
```

The output is linear in the region $f_c \pm \Delta f$. Beyond this, the detector becomes nonlinear.

Transfer characteristic (linear region):

$$V_{out}(t) = k_d \cdot [f_i(t) - f_c] = k_d \cdot k_f \cdot m(t)$$

where k_d (V/Hz) is the detector sensitivity.

Advantages:

- Simple circuit, no phase-shift network required
- Good linearity within operating range

Limitations:

- Not self-limiting - sensitive to amplitude variations (AM noise)
- Requires a limiter stage before it for proper FM demodulation
- Bandwidth limited by the individual tuned circuit bandwidths

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12

i. Illustrate the terms mean, correlation, covariance and ergodicity.

Mean (Statistical Average / Expected Value):

For a random process $X(t)$, the mean at time t is:

$$\mu_X(t) = E[X(t)] = \int_{-\infty}^{\infty} x \cdot f_X(x;t)\, dx$$

It represents the ensemble average across all realizations at time t. Physically, it is the DC level of the random process. For a WSS process, $\mu_X(t) = \mu_X$ = constant.

Example: For $X(t) = A\cos(\omega_0 t + \Theta)$ with $\Theta \sim U[0, 2\pi]$:

$$\mu_X(t) = E[A\cos(\omega_0 t + \Theta)] = 0$$

Correlation:

Autocorrelation measures similarity of a process with a time-shifted version of itself:

$$R_X(t_1, t_2) = E[X(t_1)X(t_2)]$$

For WSS: $R_X(t_1, t_2) = R_X(\tau)$, $\tau = t_1 - t_2$

$$R_X(\tau) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_1 x_2 \, f_X(x_1, x_2; \tau) \, dx_1 \, dx_2$$

$R_X(0) = E[X^2(t)]$ = total mean square power.

Cross-correlation between two processes:

$$R_{XY}(t_1, t_2) = E[X(t_1)Y(t_2)]$$

Covariance:

The autocovariance removes the mean before measuring correlation:

$$C_X(t_1, t_2) = E\left[(X(t_1) - \mu_X(t_1))(X(t_2) - \mu_X(t_2))\right]$$

$$\boxed{C_X(t_1, t_2) = R_X(t_1, t_2) - \mu_X(t_1)\mu_X(t_2)}$$

For WSS with zero mean: $C_X(\tau) = R_X(\tau)$

Variance (at $t_1 = t_2$):

$$\sigma_X^2 = C_X(0) = R_X(0) - \mu_X^2 = E[X^2] - (E[X])^2$$

Two random variables are **uncorrelated** if $C_{XY} = 0$. Independent implies uncorrelated, but not vice versa (except for Gaussian).

Ergodicity:

A process is ergodic if **time averages equal ensemble averages** for any single realization.

Mean-ergodic condition:

$$\langle X(t) \rangle = \lim_{T \to \infty} \frac{1}{2T} \int_{-T}^{T} x(t) \, dt = E[X(t)] = \mu_X$$

Correlation-ergodic condition:

$$\langle X(t)X(t+\tau) \rangle = \lim_{T \to \infty} \frac{1}{2T} \int_{-T}^{T} x(t)x(t+\tau) \, dt = R_X(\tau)$$

Practical significance: Ergodicity allows characterization of a random process from a single long observed waveform instead of requiring many ensemble samples. Most physical noise processes (e.g., thermal noise) are ergodic, which is why SNR can be measured from a single noise waveform trace. Ergodicity implies WSS, but WSS does not imply ergodicity.

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ii. Interpret the process of autocorrelation and explain the properties of autocorrelation function.

Autocorrelation Process:

The autocorrelation function (ACF) $R_X(\tau)$ quantifies the degree of statistical dependence between the values of a random process at two time instants separated by τ:

$$R_X(\tau) = E[X(t)X(t+\tau)]$$

Physically, it measures how quickly the process "remembers" or "forgets" its past values. It is computed by:

1. Taking one realization $x(t)$
2. Creating a time-shifted copy $x(t+\tau)$
3. Multiplying and taking the ensemble (or time) average

For a deterministic energy signal, the equivalent operation is the time-domain cross-correlation:

$$R_x(\tau) = \int_{-\infty}^{\infty} x(t)x(t+\tau)\,dt$$

Properties of the Autocorrelation Function:

Property 1 - Maximum at origin:

$$R_X(0) = E[X^2(t)] = \text{Total mean-square power}$$

$$|R_X(\tau)| \leq R_X(0) \quad \text{for all } \tau$$

Proof: $E[(X(t) \pm X(t+\tau))^2] \geq 0 \Rightarrow R_X(0) \pm R_X(\tau) \geq 0$

Property 2 - Even symmetry:

$$R_X(\tau) = R_X(-\tau)$$

The ACF is always an even function of τ.

Proof: $R_X(-\tau) = E[X(t)X(t-\tau)] = E[X(t+\tau)X(t)] = R_X(\tau)$

Property 3 - Periodicity:

If $X(t)$ contains a periodic component with period T_0, then $R_X(\tau)$ is also periodic with period T_0:

$$R_X(\tau + T_0) = R_X(\tau)$$

Property 4 - DC component:

If $E[X(t)] = \mu_X \neq 0$, then:

$$\lim_{\tau \to \infty} R_X(\tau) = \mu_X^2$$

This represents the DC power. For a zero-mean process, $R_X(\tau) \to 0$ as $|\tau| \to \infty$.

Property 5 - Wiener-Khinchin Theorem:

The ACF and the Power Spectral Density (PSD) form a Fourier transform pair:

$$S_X(f) = \int_{-\infty}^{\infty} R_X(\tau) e^{-j2\pi f\tau}\, d\tau \qquad \longleftrightarrow \qquad R_X(\tau) = \int_{-\infty}^{\infty} S_X(f) e^{j2\pi f\tau}\, df$$

Property 6 - Non-negative PSD:

$$S_X(f) \geq 0 \quad \text{for all } f$$

This means the ACF must be a positive semi-definite function.

Property 7 - ACF at zero lag gives total power:

$$R_X(0) = \int_{-\infty}^{\infty} S_X(f)\, df = \text{Total power}$$

Graphical interpretation:

```
R_X(τ)

R_X(0) |────────┐
       |        |\
       |        | \
μ²_X   |_ _ _ _ |_ _\_ _ _ _ _
       |            \____/
   0   └──────────────────────→ τ
       0   τ_c  (correlation time)
```

A narrow ACF (small τ_c) corresponds to a wideband PSD (fast-changing process). A wide ACF corresponds to a narrowband, slowly varying process.

13

i. Apply the theory of internal and external noise sources to analyze their impact on communication systems.

Classification of Noise:

```
                      Noise Sources
                      /            \
```

```
              External           Internal
             /    |    \         /   |   \
       Atmos. Extra-  Man-  Thermal Shot Flicker
       static terres. made
```

External Noise Sources:

1. Atmospheric (Static) Noise:

Caused by lightning discharges and other natural electrical disturbances in the atmosphere.

- PSD $\propto 1/f^2$ - falls steeply with frequency
- Most severe below 30 MHz; negligible at microwave frequencies
- Limits AM broadcast receiver performance
- Equivalent noise temperature can be thousands of Kelvin at HF

2. Extraterrestrial Noise:

- *Solar noise:* Continuous radiation from the sun, intensifies during solar flares and sunspot activity. Significant at frequencies above 10 MHz.
- *Cosmic noise:* Background radiation from galactic sources (Milky Way). Significant from 8 MHz to 1.5 GHz. Characterized by a sky noise temperature T_{sky}.
- *Cosmic Microwave Background (CMB):* $T_{CMB} \approx 2.7$ K, relevant only for very sensitive radio telescopes.

3. Industrial / Man-Made Noise:

Generated by electrical equipment: motors, ignition systems, switching supplies, fluorescent lamps, transmission lines.

- Dominant in urban areas, particularly below 600 MHz
- Modeled as impulsive noise with spectral density $\propto 1/f$

Internal Noise Sources:

1. Thermal Noise (Johnson-Nyquist Noise):

Random motion of charge carriers due to thermal energy generates voltage fluctuations across any resistor:

$$\overline{v_n^2} = 4kTBR$$

$$v_{n,rms} = \sqrt{4kTBR}$$

where $k = 1.38 \times 10^{-23}$ J/K, T = absolute temperature (K), B = noise bandwidth (Hz), R = resistance (Ω).

Available noise power from a resistor:

$$P_n = kTB$$

Equivalent circuit: noise voltage source v_n in series with a noiseless resistor R.

```
    vn = √(4kTBR)
    ┌──┤├──┬── R(noiseless) ──┐
    │      │                  │
   GND    (internal)        Terminals
```

PSD: $S_v(f) = 2kTR$ (two-sided), or $4kTR$ (one-sided) - flat (white noise).

2. Shot Noise:

Arises from the discrete nature of electron flow across a potential barrier (e.g., PN junction, vacuum tube):

$$\overline{i_n^2} = 2qI_{DC}B$$

where $q = 1.6 \times 10^{-19}$ C, I_{DC} = average DC current. Also white noise. Significant in active devices at RF frequencies.

3. Flicker Noise (1/f noise):

Due to carrier trapping/recombination at semiconductor surface states:

$$S_n(f) \propto \frac{1}{f^\alpha}, \quad \alpha \approx 1$$

Dominant at low frequencies (below 1 kHz in bipolar transistors, up to hundreds of kHz in MOSFETs). Worsens oscillator phase noise near the carrier.

4. Transit-Time Noise:

At microwave frequencies where the signal period becomes comparable to electron transit time across a device. The electron stream induces random current pulses, increasing noise significantly.

Impact on Cascaded System - Friis Formula:

For N cascaded stages with gains G_i and noise figures F_i:

$$F_{total} = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} + \cdots + \frac{F_N - 1}{\prod_{i=1}^{N-1} G_i}$$

This shows that the first stage (LNA) dominates. A high-gain, low-noise LNA suppresses the noise contribution of all subsequent stages, which is the primary design criterion for RF receiver front-ends.

---

ii. Demonstrate how noise temperature is related to noise figure by applying SNR concepts.

Noise Figure (F) - Definition from SNR:

Noise figure quantifies the degradation in signal-to-noise ratio introduced by a two-port network:

$$\boxed{F = \frac{SNR_{in}}{SNR_{out}}}$$

For an amplifier with power gain G, input signal power S_i, and input noise power $N_i = kT_0B$ (standard reference temperature $T_0 = 290$ K):

$$SNR_{in} = \frac{S_i}{kT_0B}$$

$$SNR_{out} = \frac{GS_i}{G \cdot kT_0B + N_{added,device}}$$

Therefore:

$$F = \frac{S_i/kT_0B}{GS_i/(GkT_0B + N_{device})} = 1 + \frac{N_{device}}{GkT_0B}$$

In dB: $NF = 10\log_{10}(F)$

The added noise from the device itself: $N_{device} = (F-1)GkT_0B$

Equivalent Noise Temperature (T_e):

The equivalent noise temperature models the device's internal noise as if it were produced by a hypothetical noiseless device preceded by a resistor at temperature T_e:

$$\boxed{T_e = (F-1)T_0}$$

Conversely:

$$\boxed{F = 1 + \frac{T_e}{T_0}}$$

Derivation:

Total output noise power of the amplifier:

$$N_o = G \cdot N_i + G \cdot N_{internal}$$
$$= G \cdot kT_0B + G \cdot kT_eB$$
$$= GkB(T_0 + T_e)$$

From the noise figure definition:

$$F = \frac{N_o}{GkT_0B} = \frac{GkB(T_0 + T_e)}{GkT_0B} = 1 + \frac{T_e}{T_0}$$

This confirms $T_e = (F - 1)T_0$.

SNR at output:

$$SNR_{out} = \frac{GS_i}{GkB(T_0 + T_e)} = \frac{S_i}{kB(T_0 + T_e)}$$

Or, using system noise temperature $T_{sys} = T_0 + T_e = FT_0$:

$$SNR_{out} = \frac{S_i}{kT_{sys}B}$$

Numerical relationship table:

Noise Figure (dB)	F (linear)	T_e (K)
0 dB	1.00	0 K (ideal)
1 dB	1.26	75 K
3 dB	2.00	290 K
6 dB	4.00	870 K
10 dB	10.0	2610 K

Receiver Sensitivity from SNR and Noise Figure:

Minimum detectable input signal:

$$S_{i,min} = F \cdot kT_0B \cdot SNR_{min}$$

In dBm (using $kT_0 = -174$ dBm/Hz at 290 K):

$$S_{i,min}(\text{dBm}) = -174 + NF + 10\log_{10}(B) + SNR_{min}(\text{dB})$$

Example: NF = 5 dB, B = 200 kHz, SNR_{min} = 12 dB:

$$S_{i,min} = -174 + 5 + 53 + 12 = -104\ \text{dBm}$$

This directly shows why minimizing NF improves receiver sensitivity (lower detectable signal level).

Set B

Part-A

1. Infer the need for modulation in communication systems.

Modulation is essential for practical transmission. Without it, audio signals (20 Hz - 20 kHz) would require an antenna of length $\lambda/4 \approx 3.75$ km at 20 kHz. Modulation shifts the signal to higher frequencies where antennas are small (e.g., $\lambda/4 = 75$ cm at 100 MHz). Additional benefits include: frequency-division multiplexing (multiple users share one channel), noise immunity improvements (FM offers $3\beta^2(\beta+1)$ improvement in SNR), and channel bandwidth matching.

---

2. Write and explain the general expression for an AM signal.

$$s(t) = A_c[1 + \mu\cos\omega_m t]\cos\omega_c t$$

- A_c = carrier amplitude
- $\mu = k_a A_m$ = modulation index ($0 \leq \mu \leq 1$ for no distortion)
- $\omega_m = 2\pi f_m$ = message angular frequency
- $\omega_c = 2\pi f_c$ = carrier angular frequency

Expanding: $s(t) = A_c\cos\omega_c t + \frac{\mu A_c}{2}[\cos(\omega_c+\omega_m)t + \cos(\omega_c-\omega_m)t]$

The signal contains a carrier at f_c plus sidebands at $f_c \pm f_m$. Power: $P = \frac{A_c^2}{2}\left(1 + \frac{\mu^2}{2}\right)$.

---

3. Identify whether distortion occurs if μ > 1.

Yes, severe distortion occurs when $\mu > 1$ (over-modulation). The term $[1 + \mu\cos\omega_m t]$ becomes negative during parts of the cycle, causing the envelope to go negative. An envelope detector cannot reproduce a negative envelope and instead clips, producing a distorted output with harmonics. The recovered waveform no longer resembles the original message. A limiter-type or synchronous detector is needed to handle over-modulated signals, but over-modulation still increases bandwidth due to harmonic generation.

---

4. Using Carson's rule, calculate bandwidth if Δf = 75 kHz and fm = 15 kHz.

$$B_T = 2(\Delta f + f_m) = 2(75 + 15) = \boxed{180\ \text{kHz}}$$

---

5. Classify the FM signal as NBFM or WBFM if β = 8.

Since $\beta = 8 > 1$, the signal is **Wideband FM (WBFM)**. WBFM requires multiple significant Bessel sidebands (approximately $\beta + 2 = 10$ sidebands on each side). Bandwidth $\approx 2(\beta + 1)f_m = 18f_m$. WBFM provides a significant SNR improvement of $3\beta^2(\beta + 1) = 3 \times 64 \times 9 = 1728$ (32.4 dB) over AM. Commercial FM broadcasting uses $\beta \approx 5$.

---

6. Summarize the important properties of ergodic and Gaussian random processes.

Ergodic process: Time averages equal ensemble averages (mean, autocorrelation) for any single realization. All ergodic processes are WSS. Thermal noise is a key example - its statistics can be measured from one observed waveform.

Gaussian process: The joint PDF of any set of samples is multivariate Gaussian. Completely described by mean μ_X and autocorrelation $R_X(\tau)$. Linear transformations of Gaussian processes remain Gaussian. For a Gaussian process, WSS implies SSS (the two conditions are equivalent). Uncorrelated Gaussian variables are statistically independent. Thermal noise is well-modeled as a stationary, ergodic, Gaussian process.

---

7. Outline the essential properties that an autocorrelation function must satisfy.

For a WSS process $X(t)$ with $R_X(\tau) = E[X(t)X(t + \tau)]$:

- $R_X(0) = E[X^2(t)] \geq 0$ (equals total power, always non-negative)
- $|R_X(\tau)| \leq R_X(0)$ for all τ (maximum is at origin)
- $R_X(\tau) = R_X(-\tau)$ (even symmetry)
- $R_X(\tau) \to \mu_X^2$ as $|\tau| \to \infty$ (for WSS, converges to DC power)
- $\mathcal{F}\{R_X(\tau)\} = S_X(f) \geq 0$ (Wiener-Khinchin - PSD is non-negative)

8. Apply the relationship between noise figure and noise temperature to analyze receiver characteristics.

Noise figure $F = SNR_{in}/SNR_{out}$ and equivalent noise temperature are related by:

$$T_e = (F - 1) \cdot T_0 \quad \text{and} \quad F = 1 + \frac{T_e}{T_0}$$

where $T_0 = 290$ K. A high noise figure degrades receiver sensitivity, raising the minimum detectable signal: $S_{min} = FkT_0B \cdot SNR_{min}$. For a satellite LNA with $F = 1.26$ (1 dB), $T_e = 75$ K, enabling detection of signals as weak as -130 dBm. The Friis cascade formula shows that the first stage dominates, making LNA noise figure the most critical design parameter in any receiver.

---

9. Demonstrate how thermal agitation in a resistor produces noise voltage and derive its mathematical expression.

At any temperature $T > 0$ K, free electrons in a conductor undergo random thermal motion (Brownian motion). These random collisions produce fluctuating currents with no net DC component but with a random AC component. By the fluctuation-dissipation theorem (Nyquist, 1928):

Open-circuit noise voltage:

$$\boxed{\overline{v_n^2} = 4kTBR}$$

$$v_{n,rms} = \sqrt{4kTBR}$$

where $k = 1.38 \times 10^{-23}$ J/K, T = temperature in Kelvin, B = noise bandwidth (Hz), R = resistance (Ω).

Available noise power (into matched load $R_L = R$):

$$P_n = \frac{v_n^2}{4R} \cdot R = kTB$$

This is independent of resistance - same power regardless of R. The PSD is flat (white) up to $\sim 10^{13}$ Hz (quantum effects limit it at very high frequencies).

Part-B

10

i. Construct the balanced modulator configuration for DSB-SC generation and describe its operation.

Principle:

A balanced modulator suppresses the carrier component entirely, producing Double Sideband Suppressed Carrier (DSB-SC). This is achieved by using a balanced (symmetrical) circuit configuration where carrier components cancel by phase opposition.

Ring (Lattice) Diode Modulator - Circuit:

```
                D1          D2
    ┌──────────►|────┬────|◄──────────┐
    │         (anode      cathode)    │
    │                │                │
    │       [Center-tapped T1]    [Center-tapped T2]
    │                │                │
   m(t) ─────────────┤                ├──────── DSB-SC out
   (message)         │                │
    │                │                │
    │         (cathode      anode)    │
    └─────────|◄─────┴─────►|─────────┘
                D3          D4
                      ↕
              c(t) = Ac·cos(ωct)
              [applied to center-taps of T1 and T2]
```

Operation - Positive half-cycle of c(t):

- D1 and D4 forward-biased, D2 and D3 reverse-biased
- Current path: top center-tap → D1 → load → D4 → bottom center-tap
- $m(t)$ is passed with **positive polarity** to the output transformer

Operation - Negative half-cycle of c(t):

- D2 and D3 forward-biased, D1 and D4 reverse-biased
- $m(t)$ is passed with **inverted polarity** to the output transformer

Net effect: $m(t)$ is multiplied by a square wave $p(t)$ synchronized with the carrier.

Mathematical Analysis:

The switching square wave:

$$p(t) = \frac{4}{\pi} \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} \cos[(2n+1)\omega_c t]$$

Output before filtering:

$$v_o(t) = m(t) \cdot p(t) = \frac{4}{\pi} m(t) \cos \omega_c t - \frac{4}{3\pi} m(t) \cos 3\omega_c t + \cdots$$

After BPF (passing only the term at ω_c):

$$\boxed{s_{DSB-SC}(t) = \frac{4A_m}{\pi} \cos \omega_m t \cdot \cos \omega_c t}$$

For $m(t) = A_m \cos \omega_m t$, expanding:

$$s_{DSB-SC}(t) = \frac{2A_m}{\pi} [\cos(\omega_c - \omega_m)t + \cos(\omega_c + \omega_m)t]$$

Spectrum (no carrier component):

```
S(f)
  |
  |           |           |           |
  |           ▼           |           ▼
  |      (LSB)|           |           |(USB)
──┼───────────┬───────────┬───────────┬──────→ f
             fc-fm        fc         fc+fm
                     (no spike = no carrier)
```

Carrier suppression depends on the balance of the diode pairs. Practical circuits achieve 40-50 dB carrier suppression. Demodulation requires a coherent carrier reference at the receiver (the missing carrier must be regenerated via a pilot tone or Costas loop).

Power efficiency: 100% of transmitted power is in the sidebands (none wasted on carrier), making DSB-SC power-efficient compared to conventional AM (where carrier carries $\sim 66\%$ of power at $\mu = 1$).

---

ii. Apply the concept of vestigial sideband modulation to design a transmitter and receiver block diagram and explain their operation.

Motivation for VSB:

SSB requires an ideal sharp-cutoff filter at f_c (impractical near DC). DSB uses twice the necessary bandwidth. VSB is the engineering compromise: one full sideband is transmitted along with a controlled vestige of the other, using a gradual roll-off filter.

VSB Transmitter Block Diagram:

```
                ┌──────────────────────────────┐
                │          TRANSMITTER         │
```

```
                |                              |
m(t) ──────────►[DSB-SC Modulator]────────────►[VSB Sideband Filter]──→ s_VSB(t)
                |          ↑                   |    H_VSB(f)            |
                |    Ac·cos(ωct)               |                        |
                |    [Carrier Source]          |                        |
                └──────────────────────────────┘
```

VSB Filter Response $H_{VSB}(f)$:

```
|H_VSB(f)|
  1.0 |─────────────────────┐
      |                     | \
  0.5 |                     |  ×  (transition at fc)
      |                     |   \
  0    |─────────────────────┘    ─────────────→ f
        (LSB vestige)  fc    (full USB)
        ← fv →         ←  W  →
```

The filter satisfies the Nyquist vestigial symmetry condition:

$$H_{VSB}(f_c + f) + H_{VSB}(f_c - f) = 1, \quad |f| \leq W$$

This ensures that when demodulated, the two partial sideband contributions sum to the original full spectrum.

VSB Receiver Block Diagram (Coherent Detection):

```
                ┌─────────────────────────────────┐
                |           RECEIVER              |
                |                                 |
r(t) ──────────►[BPF / IF filter]──────────────►[×]──────────►[LPF]──────→ m̂(t)
                |                                 |  ↑              |
                |                                 |  Ac·cos(ωct)    |
                |                                 |  [Coherent      |
                |                                 |   carrier]      |
                └─────────────────────────────────┘
```

Receiver Operation (coherent detection):

Received signal $r(t) = s_{VSB}(t) + n(t)$.

After BPF: Noise is limited to signal bandwidth.

After multiplying by coherent carrier $\cos \omega_c t$ and applying LPF:

The vestigial symmetry condition ensures that the partial sidebands from USB and LSB vestige sum correctly:

$$\hat{m}(t) = \frac{1}{2}[m(t) * h_{VSB}(t)] + \frac{1}{2}[m(t) * h_{VSB}(t)] \propto m(t)$$

No additional equalization is needed if the VSB filter satisfies the symmetry criterion.

Bandwidth comparison:

Modulation	Bandwidth	Carrier transmitted
DSB-AM	$2W$	Yes
DSB-SC	$2W$	No
SSB	W	No
VSB	$W + f_v$ (typ. $\sim 1.25W$)	Typically No

Application - NTSC Television:

- Video baseband: DC to 4 MHz
- VSB filter retains: 4 MHz USB + 1.25 MHz vestige of LSB
- Total video bandwidth: 5.25 MHz within a 6 MHz channel
- Sound carrier: 4.5 MHz above video carrier (FM modulated)

Advantage over SSB: Near-DC video content (scene changes, slow pans) is preserved without the phase distortion that SSB would introduce near $f = 0$. The VSB filter's gradual roll-off is much easier to realize than an ideal brick-wall SSB filter.

11

i. Show how Wideband FM is expressed mathematically and evaluate its performance characteristics relative to Narrowband FM.

Narrowband FM (NBFM) - β < 0.3:

For small β, using the approximation $\cos(\beta \sin\theta) \approx 1$ and $\sin(\beta \sin\theta) \approx \beta \sin\theta$:

$$s_{NBFM}(t) = A_c \cos\omega_c t - A_c\beta \sin\omega_m t \sin\omega_c t$$

Expanding using product-to-sum identities:

$$\boxed{s_{NBFM}(t) \approx A_c \cos\omega_c t + \frac{A_c\beta}{2}[\cos(\omega_c + \omega_m)t - \cos(\omega_c - \omega_m)t]}$$

Only three frequency components (carrier + 2 sidebands), bandwidth $\approx 2f_m$.

Wideband FM (WBFM) - Mathematical Expression:

For arbitrary β, use the Bessel function expansion of the FM phasor $e^{j\beta \sin \omega_m t}$:

$$e^{j\beta \sin \omega_m t} = \sum_{n=-\infty}^{\infty} J_n(\beta) e^{jn\omega_m t}$$

Taking the real part:

$$\boxed{s_{WBFM}(t) = A_c \sum_{n=-\infty}^{\infty} J_n(\beta) \cos(\omega_c + n\omega_m)t}$$

where $J_n(\beta)$ are Bessel functions of the first kind, order n.

Bessel coefficients $J_n(\beta)$ for key values:

n	β = 0.5 (NBFM)	β = 1	β = 5 (WBFM)
0	0.938	0.765	-0.178
1	0.242	0.440	-0.328
2	0.031	0.115	0.047
3	0.003	0.020	0.365
4	-	0.002	0.391
5	-	-	0.278

Significant sidebands (where $|J_n(\beta)| > 0.01$) extend to approximately $n_{max} \approx \beta + 2$.

Bandwidth (Carson's rule for WBFM):

$$B_{WBFM} = 2(\Delta f + f_m) = 2f_m(1 + \beta)$$

Total Power (independent of β):

$$P_{total} = \frac{A_c^2}{2} \sum_{n=-\infty}^{\infty} J_n^2(\beta) = \frac{A_c^2}{2}$$

Performance Comparison - NBFM vs WBFM:

SNR Analysis:

For FM, output SNR (post-detection) relative to input:

$$SNR_{FM,out} = \frac{3\beta^2(\beta+1)A_c^2/2}{N_0 W}$$

FM figure of merit compared to baseband:

$$\frac{SNR_{FM}}{SNR_{baseband}} = 3\beta^2(\beta+1)$$

For NBFM ($\beta = 0.5$): Improvement = $3 \times 0.25 \times 1.5 = 1.125$ (only 0.5 dB above baseband)

For WBFM ($\beta = 5$): Improvement = $3 \times 25 \times 6 = 450$ (26.5 dB above baseband)

Pre-emphasis and FM noise PSD:

FM demodulated noise has parabolic PSD:

$$S_{no}(f) = \frac{N_0 f^2}{A_c^2 k_f^2}, \quad |f| \leq W$$

This means high-frequency message components suffer more noise. Pre-emphasis (boost high frequencies before FM) and de-emphasis (attenuate at receiver) are used to equalize SNR across the audio band in commercial FM.

Full Comparison Table:

Parameter	**NBFM ($\beta < 0.3$)**	**WBFM ($\beta > 1$)**
Bandwidth	$\approx 2f_m$	$2(\Delta f + f_m)$
Spectrum	3 components	Many sidebands
SNR improvement	$\approx 3\beta^2$ (small)	$3\beta^2(\beta+1)$ (large)
Power in carrier	$A_c^2 J_0^2(\beta)/2$ (high)	Can be low (for high β)
Bandwidth efficiency	High	Low
Applications	Two-way radio, 25 kHz channels	FM broadcast, satellite links
Threshold effect	Less critical	Requires input SNR $>$ threshold

Threshold effect in WBFM: Below a threshold CNR (typically 10 dB above kT_0B), FM performance degrades rapidly ("FM clicks"). WBFM trades bandwidth for SNR improvement, valid only above threshold.

---

ii. Construct the phasor representation of voltages in a Foster-Seeley discriminator and explain how it recovers the modulating signal.

Circuit Configuration:

The Foster-Seeley discriminator uses coupled resonant circuits and exploits the phase relationship between primary and secondary voltages to detect frequency deviation.

```
                 RFC (Radio Frequency Choke)
            ┌──UUUUUU──┬───────────────────┐
            │          │                   │
 Vi ───[Lp, Cp]      [Choke                 │
 (Primary   (primary   bypasses RF         │
  tank)      tuned     to midpoint)        │
             to fc)                        │
                       ┌──Ls/2──[A]──[D1]──┴───[R1||Ca]──┐
                       │                                 │
                       ┤ (mutual inductance M)           ├── Vout
                       │                                 │
                       └──Ls/2──[B]──[D2]──┬───[R2||Cb]──┘
                                           │
                                          GND
```

Key: The primary voltage V_1 is fed to the center of the secondary via the RFC choke. Thus:

- Voltage at A: $V_A = \frac{V_2}{2} + V_1$ (vectorially)
- Voltage at B: $V_B = -\frac{V_2}{2} + V_1$ (vectorially)

where V_2 is the secondary induced voltage.

Phasor Analysis:

At resonance ($f = f_c$):

- Mutual coupling voltage: $V_2 \propto j\omega M I_1$ → secondary current I_2 lags V_1 by 90°
- V_2 is perpendicular (90°) to V_1

```
          V2/2 (↑ perpendicular to V1)
           |
      VA = V1 + V2/2 (equal magnitude to VB)
           |
──────────┼──────────────→  V1 (reference)
           |
      VB = V1 - V2/2
           |
          -V2/2 (↓)

|VA| = |VB|  →  V_D1 = V_D2  →  Vout = 0
```

Above resonance ($f > f_c$), secondary becomes inductive, phase shift < 90°:

```
      V2/2 ↗ (tilts toward V1)
         /
───────/────────────→ V1
  VA /  (longer diagonal)
    /
   VB (shorter diagonal, V2/2 tilts away)

|VA| > |VB|  →  Vout > 0
```

Below resonance ($f < f_c$), secondary becomes capacitive, phase shift > 90°:

```
      V2/2 ↗ (tilts away from V1)
         \
───────\────────────→ V1
  VA \ (shorter diagonal)
      \
       VB (longer diagonal)

|VA| < |VB|  →  Vout < 0
```

Output Voltage:

$$V_{out} = |V_A| - |V_B|$$

$$= \sqrt{V_1^2 + \left(\frac{V_2}{2}\right)^2 + V_1\frac{V_2}{2}\cos\phi} - \sqrt{V_1^2 + \left(\frac{V_2}{2}\right)^2 - V_1\frac{V_2}{2}\cos\phi}$$

where ϕ is the phase angle between V_1 and $V_2/2$.

At $f = f_c$: $\phi = 90^\circ$, $\cos\phi = 0$, $V_{out} = 0$

For small deviations: $V_{out} \approx k_d(f - f_c) = k_d k_f m(t)$ (linear discriminator output)

S-Curve Characteristic:

```
Vout
(V)
+Vp |          ___________
    |         /
    |        / (linear region)
  0 |───────/─────────────────→ f
    |      /     fc
    |     /
-Vp |____/
```

```
  ← fv →
 linear range
```

The slope of the linear portion (k_d in V/Hz) determines the discriminator gain. The linear range spans approximately $\pm\Delta f_{max}$ around f_c.

Signal recovery:

Since $f_i(t) = f_c + k_f m(t)$:

$$V_{out}(t) = k_d[f_i(t) - f_c] = k_d k_f m(t)$$

The original message $m(t)$ is recovered with amplitude $k_d k_f$.

Comparison with Balanced Slope Detector:

Feature	Balanced Slope	Foster-Seeley
Linearity	Poor (uses two separate resonant slopes)	Better (uses phase variation)
Sensitivity	Lower	Higher
Bandwidth	Narrower	Wider
Limiter needed	Yes	Yes
Component count	More (two tuned circuits)	Single coupled pair

Both require a limiter before them. The ratio detector is a modification of Foster-Seeley that provides inherent amplitude limiting.

12

i. Formulate the mathematical representation of a random process using statistical parameters.

Definition:

A random process $\{X(t, \zeta) : t \in T, \zeta \in S\}$ is a two-dimensional function:

- For fixed ζ_k: $X(t, \zeta_k) = x_k(t)$ is the k-th sample realization (deterministic waveform)
- For fixed t_k: $X(t_k, \zeta)$ is a random variable

The ensemble is the collection of all realizations $\{x_1(t), x_2(t), \ldots, x_N(t)\}$.

First-Order Statistical Description:

First-order CDF:

$$F_X(x;t) = P[X(t) \le x]$$

First-order PDF:

$$f_X(x;t) = \frac{\partial F_X(x;t)}{\partial x}$$

Mean (First Moment):

$$\mu_X(t) = E[X(t)] = \int_{-\infty}^{\infty} x \cdot f_X(x;t)\, dx$$

Mean-Square Value:

$$E[X^2(t)] = \int_{-\infty}^{\infty} x^2 f_X(x;t)\, dx$$

Variance:

$$\sigma_X^2(t) = E[X^2(t)] - \mu_X^2(t) = E[(X(t) - \mu_X)^2]$$

Second-Order Statistical Description:

Joint PDF at two time instants t_1, t_2:

$$f_X(x_1, x_2; t_1, t_2) = \frac{\partial^2 F_X(x_1, x_2; t_1, t_2)}{\partial x_1 \partial x_2}$$

Autocorrelation Function:

$$R_X(t_1, t_2) = E[X(t_1)X(t_2)] = \iint x_1 x_2 \cdot f_X(x_1, x_2; t_1, t_2)\, dx_1\, dx_2$$

Autocovariance:

$$C_X(t_1, t_2) = R_X(t_1, t_2) - \mu_X(t_1)\mu_X(t_2)$$

For WSS process ($\tau = t_1 - t_2$):

$$R_X(\tau) = E[X(t)X(t+\tau)]$$

Power Spectral Density (Wiener-Khinchin):

$$\boxed{S_X(f) = \int_{-\infty}^{\infty} R_X(\tau) e^{-j2\pi f\tau}\, d\tau}$$

Total power: $P_X = R_X(0) = \int_{-\infty}^{\infty} S_X(f)\, df$

Example: Random Sinusoidal Process

$$X(t) = A\cos(\omega_0 t + \Theta), \quad \Theta \sim U[0, 2\pi]$$

Mean:

$$\mu_X(t) = E[A\cos(\omega_0 t + \Theta)] = \frac{A}{2\pi}\int_0^{2\pi} \cos(\omega_0 t + \theta)\, d\theta = 0$$

Autocorrelation:

$$R_X(t, t+\tau) = E[A^2 \cos(\omega_0 t + \Theta)\cos(\omega_0 t + \omega_0 \tau + \Theta)]$$
$$= \frac{A^2}{2}\cos\omega_0\tau = R_X(\tau) \quad \text{(function of } \tau \text{ only)}$$

This process is **WSS**: constant mean and τ-dependent autocorrelation.

PSD:

$$S_X(f) = \mathcal{F}\left\{\frac{A^2}{2}\cos\omega_0\tau\right\} = \frac{A^2}{4}[\delta(f - f_0) + \delta(f + f_0)]$$

This shows discrete power at $\pm f_0$.

Response of LTI System to a Random Process:

For WSS input $X(t)$ through LTI system $H(f)$:

$$S_Y(f) = |H(f)|^2 S_X(f)$$
$$\mu_Y = \mu_X \cdot H(0)$$
$$R_Y(\tau) = \mathcal{F}^{-1}\{|H(f)|^2 S_X(f)\}$$

This result (valid only for WSS input) is fundamental to noise analysis in communication receivers.

---

ii. Show how the statistical properties determine whether a process is WSS or SSS and compare them.

Strict-Sense Stationary (SSS):

A process is SSS (or strongly stationary) if its complete joint statistical description is invariant to time translation. For any n, any set $\{t_1, t_2, \ldots, t_n\}$, and any time shift ϵ:

$$f_X(x_1, x_2, \ldots, x_n;\, t_1, t_2, \ldots, t_n) = f_X(x_1, x_2, \ldots, x_n;\, t_1 + \epsilon, t_2 + \epsilon, \ldots, t_n + \epsilon)$$

This means **all statistical moments** (mean, variance, higher-order moments, and all joint distributions) are shift-invariant.

Consequences:

- Mean: $E[X(t)] = \mu$ (constant)
- Autocorrelation: $R_X(t_1, t_2) = R_X(\tau)$
- Third moment: $E[X^3(t)] = $ constant
- All higher moments: time-invariant

Wide-Sense Stationary (WSS):

A weaker condition - only requires the first and second moments to be shift-invariant:

WSS Conditions:

1. $E[X(t)] = \mu_X$ (constant, independent of t)
2. $R_X(t_1, t_2) = R_X(\tau)$, where $\tau = t_1 - t_2$

WSS does **not** require higher-order statistics to be stationary.

Determination using Statistical Properties:

To check **WSS**:

1. Compute $\mu_X(t)$. If it varies with t, **not WSS**.
2. Compute $R_X(t_1, t_2)$. If it depends on both t_1 and t_2 (not just $\tau = t_1 - t_2$), **not WSS**.

Example test (WSS):

Process: $X(t) = A(t)\cos(\omega_0 t)$ where $A(t)$ has mean μ_A and $R_A(\tau)$.

$$\mu_X(t) = \mu_A \cos\omega_0 t \quad \leftarrow \text{varies with } t\text{: NOT WSS}$$

To check **SSS**: Must verify invariance of all joint PDFs under time shifts - practically intractable for most processes except Gaussian.

Comparison Table:

Aspect	WSS	SSS
Conditions	2 conditions (mean + ACF)	Infinite-order joint PDF invariance
Strength	Weak (necessary)	Strong (sufficient)

Aspect	WSS	SSS
Practical verification	Feasible	Usually intractable
Implication	SSS $\Rightarrow$ WSS (always)	WSS $\Rightarrow$ SSS only for Gaussian
All moments stationary?	No (only 1st and 2nd)	Yes
PSD valid?	Yes (via Wiener-Khinchin)	Yes
Higher-order spectra	Not guaranteed	Valid
Example	WSS but not SSS: non-Gaussian process with stationary mean and ACF	White Gaussian Noise, thermal noise

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Key Result for Gaussian Processes:

For a **Gaussian** random process:

$$\text{WSS} \iff \text{SSS}$$

This equivalence holds because a Gaussian process is entirely characterized by its mean and covariance function. If these are shift-invariant (WSS conditions), then the entire probability structure (all joint PDFs) is automatically shift-invariant (SSS).

This equivalence is why Gaussian noise is so tractable - we only need to verify the two WSS conditions to conclude full stationarity.

Impact on LTI System Analysis:

WSS is the minimum requirement for valid frequency-domain noise analysis:

$$S_Y(f) = |H(f)|^2 S_X(f)$$

This relation holds only if $X(t)$ is WSS. SSS is stronger than needed - WSS suffices for all linear system noise performance calculations in communications.

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i. Show how noise figure influences overall receiver sensitivity and performance.

Noise Figure and Sensitivity - Definitions:

Noise Figure:

$$F = \frac{SNR_{in}}{SNR_{out}} = \frac{S_i/N_i}{S_o/N_o} \geq 1$$

Receiver Sensitivity is the minimum input signal power $S_{i,min}$ that produces a specified output SNR_{min}:

With input noise $N_i = kT_0B$:

$$S_{i,min} = F \cdot kT_0B \cdot SNR_{min}$$

In logarithmic form (using $kT_0 \approx -174$ dBm/Hz at 290 K):

$$\boxed{S_{i,min}(\text{dBm}) = -174 + NF(\text{dB}) + 10\log_{10} B + SNR_{min}(\text{dB})}$$

This equation directly shows that every 1 dB increase in noise figure raises the minimum detectable signal by 1 dB - degrading receiver sensitivity by 1 dB.

Cascaded Receiver Chain Analysis (Friis Formula):

For a practical receiver chain:

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Antenna → [LNA, G1, F1] → [Mixer, G2, F2] → [IF Amp, G3, F3] → [Detector]
```

Total noise figure:

$$F_{total} = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1G_2} + \cdots$$

Critical observation: If $G_1 >> 1$ (high gain LNA), the contributions from $F_2, F_3, \ldots$ become negligible. The LNA dominates system noise.

Numerical example:

Stage	Gain	NF
LNA	20 dB (G=100)	2 dB (F=1.58)
Mixer	-6 dB (G=0.25)	10 dB (F=10)
IF Amp	30 dB (G=1000)	6 dB (F=4)

$$F_{total} = 1.58 + \frac{10-1}{100} + \frac{4-1}{100 \times 0.25} = 1.58 + 0.09 + 0.12 = 1.79$$

$$NF_{total} = 10\log(1.79) = 2.53 \text{ dB}$$

Despite the noisy mixer (10 dB NF), the high-gain LNA keeps total NF near 2.5 dB.

Without LNA (mixer first):

$$F_{total} = 10 + \frac{4-1}{0.25} = 10 + 12 = 22 \quad (13.4 \text{ dB})$$

This would degrade sensitivity by $13.4 - 2.53 \approx 11$ dB - a huge degradation.

Sensitivity floor calculations:

Application	NF (dB)	B (Hz)	SNR_min (dB)	Sensitivity (dBm)
FM radio	10	200 kHz	12	-99
GSM cellular	8	200 kHz	9	-110
GPS	2	2 MHz	14	-135
Satellite LNA	0.5	36 MHz	10	-108

Noise Figure improvement techniques:

1. **LNA placement first:** Always place lowest NF stage at the antenna
2. **Minimize cable/filter loss** before LNA: Every 1 dB of loss = 1 dB added to NF
3. **Cryogenic cooling:** Reduces T_e (used in radio telescopes, $T_e < 10$ K)
4. **GaAs HEMT LNAs:** Achieve NF of 0.3-0.5 dB at microwave frequencies

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ii. Demonstrate how narrowband noise is represented and analyze the behavior of its in-phase and quadrature components.

Narrowband Noise - Definition:

Any broadband noise $n(t)$ (e.g., white noise $S_n(f) = N_0/2$) passed through a bandpass filter (BPF) of bandwidth $B << f_c$ becomes narrowband noise.

Canonical Representation:

Any narrowband noise process with center frequency f_c can be uniquely written as:

$$\boxed{n(t) = n_I(t)\cos(2\pi f_c t) - n_Q(t)\sin(2\pi f_c t)}$$

Or equivalently in envelope-phase form:

$$n(t) = r(t)\cos[2\pi f_c t + \psi(t)]$$

where:

- $n_I(t) = $ In-phase component (lowpass)
- $n_Q(t) = $ Quadrature component (lowpass)
- $r(t) = \sqrt{n_I^2(t) + n_Q^2(t)}$ = envelope (Rayleigh distributed if zero-mean Gaussian)
- $\psi(t) = \arctan\left(\frac{n_Q(t)}{n_I(t)}\right)$ = phase (uniform distributed)

Extraction of components:

$$n_I(t) = [n(t)\cos(2\pi f_c t)] * h_{LP}(t) \times 2$$
$$n_Q(t) = -[n(t)\sin(2\pi f_c t)] * h_{LP}(t) \times 2$$

Statistical Properties of $n_I(t)$ and $n_Q(t)$:

For narrowband Gaussian noise with PSD $S_n(f)$ and bandwidth B:

1. Identical PSD:

$$S_{n_I}(f) = S_{n_Q}(f) = \begin{cases} S_n(f - f_c) + S_n(f + f_c), & |f| \leq B/2 \\ 0, & \text{otherwise} \end{cases}$$

For white noise through BPF: $S_{n_I}(f) = S_{n_Q}(f) = N_0$ for $|f| \leq B/2$.

2. Equal Power:

$$E[n_I^2(t)] = E[n_Q^2(t)] = E[n^2(t)] = N_0 B$$

The variance of each component equals the total noise power.

3. Uncorrelated (and independent for Gaussian):

$$E[n_I(t) n_Q(t)] = 0$$

At the same time instant, n_I and n_Q are orthogonal. If $n(t)$ is Gaussian, they are statistically independent.

4. Both are lowpass with bandwidth $B/2$.

5. Gaussian: If $n(t)$ is Gaussian, n_I and n_Q are jointly Gaussian.

Behavior in AM Demodulation (Envelope Detector):

Received signal: $r(t) = [A_c + m(t)]\cos\omega_c t + n(t)$

Expanding narrowband noise:

$$r(t) = [A_c + m(t) + n_I(t)] \cos \omega_c t - n_Q(t) \sin \omega_c t$$

Envelope:

$$R(t) = \sqrt{[A_c + m(t) + n_I(t)]^2 + n_Q^2(t)}$$

High SNR ($A_c >> n$):

$$R(t) \approx A_c + m(t) + n_I(t)$$

The quadrature component $n_Q(t)$ is suppressed. Output noise is just $n_I(t)$, with power $N_0 B$.

$$SNR_{AM} = \frac{\mu^2 A_c^2/2}{N_0 W} \quad \text{(for tone modulation, } W = f_m\text{)}$$

Behavior in FM Demodulation:

The FM discriminator responds to instantaneous frequency. For received signal $r(t) = s_{FM}(t) + n(t)$:

Instantaneous phase of $r(t)$ (high SNR approximation):

$$\phi_i(t) \approx \phi_{FM}(t) + \frac{n_Q(t)}{A_c}$$

Discriminator output (differentiates instantaneous phase):

$$v_o(t) = \frac{1}{2\pi} \frac{d}{dt} \left[\phi_{FM}(t) + \frac{n_Q(t)}{A_c} \right] = k_f m(t) + \frac{1}{2\pi A_c} \frac{dn_Q(t)}{dt}$$

FM output noise PSD (differentiation corresponds to $\times j2\pi f$ in frequency domain):

$$S_{no,FM}(f) = \frac{(2\pi f)^2}{(2\pi A_c)^2} S_{n_Q}(f) = \frac{f^2 N_0}{A_c^2}$$

This **parabolic (f²) noise spectrum** is the defining characteristic of FM demodulated noise - high-frequency message components are noisier than low-frequency ones.

```
S_no(f)
 |                    /
 |                   /
 |                  /  (f² shaped)
 |                 /
 |                /
 |_______________/
 └────────────────────→ f
  0              W
```

$$SNR_{FM} = \frac{3\beta^2(\beta+1)A_c^2/2}{N_0 W} = 3\beta^2(\beta+1) \cdot SNR_{baseband}$$

Rayleigh envelope distribution (no signal):

$$f_r(r) = \frac{r}{\sigma^2} e^{-r^2/2\sigma^2}, \quad r \geq 0$$

where $\sigma^2 = N_0 B$ is the noise variance. This is the distribution of the noise envelope seen by an AM envelope detector in the absence of signal.